

Generation Interconnection Request System Impact Study Report

For Queue Project Q11-0055

Sabanera 500 kV
July 28, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0055. The Interconnection Customer proposes a 844.0 MW natural gas combined cycle facility interconnecting at the Sabanera 500 kV substation in the Indiana area. The request was received on July 10, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 43 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$75,356,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of SWAN VALLEY - TREGO 500 kV, the TWIN COUNTY RANCH - SWAN VALLEY 230 kV loads to 112.8% of its 439 MVA rating. Project DFAX: 11.9%.

MONFORT - WABASH 230 kV line: 103.9% loading (818 MVA) following the outage of the MONTPELIER ELECTRIC GENERATING STATION SUBSTATION - WABASH 500 kV; DFAX = 11.7%.

SBANERA 500-230 kV Auto: 101.3% loading (969 MVA) following the outage of the SABANERA - LONETREE 500 kV; DFAX = 40.1%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SWAN VLY - TWIN COUNTY RANCH 230 kV	Rebuild line with higher-capacity conductor	\$40,969,000	100% (\$40,969,000)
MONFORT - WABASH 230 kV Line	Reconductor line	\$12,887,000	100% (\$12,887,000)
SABANERA 500/230kV Transformer Bank	Replace transformer with higher-rated unit	\$21,500,000	100% (\$21,500,000)
	Total allocated to project		\$75,356,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.